Risk-Return Tradeoff

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1 Introduction

A stock currently priced at \$1,000. We evaluate the expected return and risk (volatility) based on three discrete scenarios: Worst Case, Base Case, and Best Case.

2 Scenario Definitions

The following table summarizes the projected prices and their associated probabilities. The return (R_i) is calculated as $(P_i - P_0)/P_0$.

Scenario	**Target Price (P_i)**	**Return (R_i)**	**Probability (P_i)**
Worst Case	\$900	-10.0%	30%
Base Case	\$1,050	+5.0%	50%
Best Case	\$1,300	+30.0%	20%

Table 1: Stock Price Scenarios

3 Quantitative Calculations

3.1 Expected Return

The expected return $E[R]$ is the probability-weighted average of all possible returns:

$$E[R] = \sum_{i=1}^{n} (P_i \times R_i)$$

$$\begin{aligned}
E[R] &= (0.30 \times -0.10) + (0.50 \times 0.05) + (0.20 \times 0.30) \\
E[R] &= -0.03 + 0.025 + 0.06 \\
E[R] &= 0.055 \text{ or } \mathbf{5.5}\%
\end{aligned}$$

3.2 Risk (Variance and Standard Deviation)

Variance (σ^2) measures the dispersion of returns around the mean:

$$\sigma^2 = \sum_{i=1}^{n} P_i \times (R_i - E[R])^2$$

$$\text{Worst Case component: } 0.30 \times (-0.10 - 0.055)^2 = 0.0072075$$
$$\text{Base Case component: } 0.50 \times (0.05 - 0.055)^2 = 0.0000125$$
$$\text{Best Case component: } 0.20 \times (0.30 - 0.055)^2 = 0.012005$$
$$\sigma^2 = 0.019225$$

Standard Deviation (σ) is the square root of the variance:

$$\sigma = \sqrt{0.019225} \approx \mathbf{13.87}\%$$

4 Analysis of Risk-Return Trade-off

- **Expected Return:** 5.5%
- **Volatility (Risk):** 13.87%
- **Coefficient of Variation (CV):** $\frac{\sigma}{E[R]} = \frac{13.87}{5.5} \approx 2.52$

The investment exhibits a CV of 2.52, indicating that for every 1% of expected return, the investor assumes 2.52% of risk. The high volatility is driven primarily by the "Best Case" outlier (+30%) and the significant downside probability (30% chance of a 10% loss).